UNIT 3 · TOPIC 6(B)

Application-Rate Calculations: Seed, Fertilizer and Chemicals

Syllabus topic: Calculations of field capacity, efficiency and rates of seed, fertilizer and chemical applicators

The calibration procedures are in Topic 1; this file gathers the rate formulas with worked numbers.

Seed drill

$$A\ (m^2) = \pi D \times N \times W;\ \text{seed rate (kg/ha)} = 10{,}000 \times \text{weight (kg)} \div A\ (m^2);\ N \text{ for area } A = A/(\pi DW)$$

Ex.1 9-row drill, row spacing 20 cm (W = 1.8 m), ground wheel D = 1.2 m, wheel turned N = 25 rev; seed collected = 1.05 kg. A = π × 1.2 × 25 × 1.8 = **169.6 m²**. Seed rate = 10,000 × 1.05/169.6 ≈ **62 kg/ha**. If the target is 100 kg/ha for wheat, increase the exposed flute length and repeat until within tolerance.

Ex.2 (revolutions for 1/25 ha) N = 400/(π × 1.2 × 1.8) = 400/6.79 ≈ **59 revolutions**.

Planter

$$\text{Population (plants/ha)} = 10{,}000 \div [\text{row spacing (m)} \times \text{seed spacing (m)}]$$

Ex. Maize at 75 × 20 cm → 10,000/(0.75 × 0.20) = **66,667 seeds/ha**; at 75 × 25 cm → **53,333/ha**.

Sprayer and duster

$$Q\ (L/ha) = 600 \times q_total\ (L/min) \div [v\ (km/h) \times w\ (m)];\ \text{duster rate (kg/ha)} = 600 \times (kg/min) \div (v \times w)$$

Ex. 12 nozzles × 0.8 L/min = 9.6 L/min; v = 5 km/h, w = 6 m → Q = 600 × 9.6/(5 × 6) = **192 L/ha**.

Speed from a timed run. 30 m in 21.6 s → 30/21.6 = 1.389 m/s × 3.6 = **5 km/h**.

Applicator work rate

Ex.3 A sprayer applies 200 L/ha with a 600-L tank at EFC 2 ha/h → one tank lasts 3 ha = 1.5 h; refills per 8-h day ≈ 5; chemical handled = **3200 L/day**.

Tank loads. 12 ha × 200 L/ha = 2400 L ÷ 600 L = **4 tank loads**.

- 1 ha = 10,000 m².

Practice MCQs

Q3. A 9-row drill (20-cm spacing, W = 1.8 m) with 1.2-m wheel turned 25 revolutions covers about:

(a) 68 m² (b) 170 m²
(c) 340 m² (d) 425 m²

Q4. If 1.05 kg seed is collected over 169.6 m², the seed rate is about:

(a) 42 kg/ha (b) 62 kg/ha
(c) 84 kg/ha (d) 105 kg/ha

Q5. Revolutions needed to simulate 1/25 ha (400 m²) with D = 1.2 m, W = 1.8 m:

(a) 25 (b) 40
(c) 59 (d) 118

Q13. Plant population for 75 × 20 cm spacing is:

(a) 33,333/ha (b) 53,333/ha
(c) 66,667/ha (d) 75,000/ha

Q14. For 75 × 25 cm spacing, population is:

(a) 43,555/ha (b) 53,333/ha
(c) 66,667/ha (d) 80,000/ha

Q20. Twelve nozzles of 0.8 L/min at 5 km/h with 6-m swath give:

(a) 96 L/ha (b) 148 L/ha
(c) 192 L/ha (d) 240 L/ha

Q21. A sprayer covers 30 m in 21.6 s. Its speed is:

(a) 3 km/h (b) 4 km/h
(c) 5 km/h (d) 6 km/h

Q33. Tank loads needed to spray 12 ha at 200 L/ha with a 600-L tank:

(a) 2 (b) 3
(c) 4 (d) 6

Answers

Q3 — (b) 170 m² — *$A = \pi DNW = 3.1416 \times 1.2 \times 25 \times 1.8 = 169.6 \approx 170$ m².*

Q4 — (b) 62 kg/ha — *Rate = 10,000×1.05/169.6 = 61.9 ≈ 62 kg/ha.*

Q5 — (c) 59 — *$N = 400/(\pi \times 1.2 \times 1.8) = 400/6.786 = 58.9 \approx 59$ rev.*

Q13 — (c) 66,667/ha — *Population = 10,000/(0.75×0.20) = 66,667.*

Q14 — (b) 53,333/ha — *Population = 10,000/(0.75×0.25) = 53,333.*

Q20 — (c) 192 L/ha — *Q = 600×(12×0.8)/(5×6) = 600×9.6/30 = 192 L/ha.*

Q21 — (c) 5 km/h — *v = 30/21.6 = 1.389 m/s × 3.6 = 5 km/h.*

Q33 — (c) 4 — *12 ha × 200 L/ha = 2400 L ÷ 600 L = 4 tank loads.*

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 11 (Crop Planting) & Ch. 21 (Spraying and Dusting); Jagdishwar Sahay — Elements of Agricultural Engineering; ISI/BIS & RNAM test codes for seed drills and sprayers; NPTEL Farm Machinery lectures. Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 2 (Field Capacities) & Ch. 17; RNAM test codes (combine, thresher, transplanter); BIS standards for power threshers and chaff cutters; Jagdishwar Sahay; NPTEL lectures.